



PEAS Scheme of Work: Chemistry

SENIOR 1

SUBJECT: Chemistry Class: S.1

Term: ONE

Teacher's Name:

Time allocation: 3 periods

When it comes to planning your lessons use the structure: starter, I do, we do, you do, plenary. The starter is a written task that reviews previous learning. Ensure your lessons provide regular and extended opportunities for independent practice.

YPR:

Y= yes, I taught the lesson

P= I partially taught it e.g. I didn't get through all the content.

R= I taught the lesson, but I think students would benefit from a review

Week Subtopic:	Learning out comes	methodology	Teaching materials	Y P R
Theme: introduction to chemistry and experimental techniques. Competency: The learner assesses the application of chemistry in our daily life and its contribution to the economy.				
1.1	Learners should be able to; Identify the common materials used in our daily life and where they are produced.	- Teacher explains and displays a chart with some common materials. - Using teacher's explanation and the relevant textbooks, Learners discuss in groups to Identify the common materials used in our daily life and where they are produced groups under the guidance of teacher and present their findings. - Learners are tasked to write a message to inform parents about the common materials used in our daily life and where they are produced.	RE star page 2 Baroque page 2	

1.2 Nature of Chemistry	The learner should be able to: Define the term chemistry	- Reviews primary science in relation to chemistry and performs an experiment to drive to the definition for example by heating substances (matter) such as plastics, papers, melting ice. - Using teacher's explanation and the relevant textbooks, Learners discuss in groups the changes that have taken place during the experiment in their respective groups under the guidance of teacher and present their findings. - Each Learner write the definition of chemistry using the relevant textbooks and also after the harmonization of the teacher.	(LSC BOOK 1 Baroque Pg 2, star Pg 2 Plastics cups, plates, bottles Heat source Water papers ice	
1.3	State and Explain why do we study chemistry	- Teacher briefly explains by telling learners how people have benefited from chemistry. - Using teacher's explanation and the relevant textbooks, Learners discuss to explain why we study chemistry under the guidance of teacher and present their findings by writing on the chalk board. - Learners are tasked to write a message to inform parents about the reasons to why we study chemistry.	RE star page 3 Textbooks	
2.1	Find out the relationship between chemistry and other subjects	Gives and explains some topics in physics and mathematics which require the knowledge of chemistry. For example, matter is taught in chemistry and physics. Accurate measurements in chemistry	RE star page 5 Chart with different subjects	

		requires the knowledge of Physics and maths. - Using teacher's explanation and the relevant textbooks Learners in groups discuss the relationship between chemistry and agriculture, biology under the guidance of a teacher and after they present their findings on the chalk board. - Learners are given task to research on the subjects involved during mining of minerals using the relevant textbooks.		
2.2	Find out the importance of chemistry in our everyday life	Teacher explains and show learners substances used in our everyday life. - Using teacher's explanation and the relevant textbooks Learners in groups discuss to find out the importance of chemistry in our everyday life under the guidance of a teacher and after they present their findings on the chalk board. - The learners write down their own notes in their exercise books using the relevant textbooks	Fertilizers Food soap and detergents drugs	
2.3	Identify careers related to the study of chemistry	- Explains the term careers. - Teacher shows some of the products got from knowledge of chemistry. - Using teacher's explanation and the relevant textbooks Learners in groups discuss to state other products and suggests careers from chemistry in their respective groups as the teacher monitors and there after they present their findings by writing on the chalk board.	RE star page 7	

		The learners write down their own notes in their exercise books using the relevant textbooks		
3.1 Nature of Chemistry	Know the contribution of chemistry to the economy of Uganda	- Explains the key term, economy. - Using teacher's explanation and the relevant textbooks Learners in groups discuss the contribution of chemistry to the economy of Uganda in groups under the guidance of the teacher and there after they present their findings by writing on the chalk board. - Each learners write short report to identify areas in chemistry which contributes to the economy of Uganda using the relevant textbooks	RE star page 8 Fertilizers Food, soap and detergents Drugs, plastics, petroleum	
3.2 Activity of integration	Ability to exhibit knowledge, skills and application of the key concepts on the nature of chemistry	Gives and guides learners on the activity of integration. Learner attempts the activity	RE; Baroque Pg 10, Star Pg 9	
Competency: The learner understands that chemistry is a process of evidence-based enquiry involving the collection of evidence about the natural world				

3.3 Experimental chemistry.	Define laboratory and state the laboratory rules and regulations.	- Tells a story about S.1 students who went in the hospital to visit their fellow friend who was sick. when they reached, they were directed to the laboratory where tests and experiments were being performed. - Using knowledge from the story and the relevant textbooks, learners in groups brainstorm to define laboratory and state the rules and regulations of the laboratory under the guidance of a teacher and there after they present their findings by writing on the chalk board - Each learner is tasked to write a report to the teacher informing him/her about the laboratory rules and regulation using the relevant textbooks	LSC baroque Pg 14-15 ,Star Pg 13-14 Charts showing laboratory rules and regulations.	
4.1	Identify the risks in the laboratory and ways of avoiding them	- Narrates a story on an accident that happened in the laboratory to S.3 student during the practical lesson. - Displays chart and reagent bottles showing hazard symbols. - Using knowledge from the story and the relevant textbooks Learners brainstorm the risks in the laboratory and ways of avoiding them under the guidance of a teacher and there after they present their findings by writing on the chalk board. - Learners are given a task to write a report about the risks and how to avoid them using the relevant textbooks.	Lsc Baroque Pg 16-18,Star Pg 13-14 Chart showing hazard symbols Videos	
4.2	Define laboratory apparatus.	Explains that the equipment's which are kept in the kitchen are called kitchenware	RE star page 17-20 Measuring cylinder Graduated beaker	

	Identify the name and use of some of the laboratory equipment (such as burette, pipette, measuring cylinders, thermometers, Bunsen burner and balance test tubes, beaker spatula pipette conical flask, flat bottom med flask gas jars, delivery tube, tripod stand, test tube holders, Bunsen burner, wire gauze)	- Using teacher's explanation and the relevant textbooks Learners in groups brainstorm about the definition of laboratory apparatus in their group sand also identify the names and uses of some apparatus under the guidance of a teacher and there after they present their findings by writing on the chalk board - Each Learners identify and draw some common apparatus using the relevant textbooks	Dropper Volumetric flask Burette Funnel Retort stand test tubes , beaker spatula pipette conical flask, flat bottom med flask gas jars, delivery tube, tripod stand, test tube holders, Bunsen burner, wire gauze,	
4.3	- Name the heating apparatus in the laboratory. - Draw a well labelled diagram of the Bunsen burner. - Explain how to use the Bunsen burner	- Teacher show the leaners Bunsen burners and explains the uses of different parts of the Bunsen burner - Using teacher's explanation and the relevant textbooks Learners in groups draw a diagram of the Bunsen burner and also Explain how to use the Bunsen burner under the guidance of the teacher - Each leaner is tasked to draw a well labelled diagram of a Bunsen burner	RE star page 27 -28 Charts to be given to groups for drawing Textbooks Bunsen burners Gas cylinder Delivery tube Wooden split	
5.1	- Define a flame - State the types of flames - give the differences between the types of flames	- teacher lights two Bunsen burners and explains to learners. - Using teacher's explanation and the relevant textbooks, Learners in groups discuss the definition of flame, the types of flames and the differences between the types of flames and also to Explain how to choose proper heating equipment	RE star page 29-30 Textbooks Two lighted Bunsen burners	

	Explain how to choose proper heating equipment	under the guidance of teacher and then after learners presents their findings by writing on the chalk board. Each learner is tasked to write notes in the exercise books using relevant textbooks.		
5.2 Experimental chemistry:	Learners should be able to understand the scientific method to carry out investigations.	- Explain the key term scientific method and illustrate the extraction of an orange juice from an orange, - Learners perform experiments of extracting juice from passions under the guidance of teacher in groups. - Learners write down the notes in their exercise books	LSC Baroque Pg 23 ,Star Pg 30 Passion fruits Water Sieve/cloth and filter funnel containers	
5.3 Purification of substances	Explain how sand can be obtained from water using decantation. Explain how sand can be obtained from water using filtration	- teacher briefly explains the term decantation - Using teacher's explanation and the relevant textbooks Learners in groups discuss and perform experiment to show how sand can be obtained from water under the guidance of the teacher and then after learners presents their findings by writing on the chalk board. - Each learner is tasked to write the notes in the exercise books using the relevant textbooks - teacher briefly explains the term filtration - Using teacher's explanation and the relevant textbooks Learners in groups discuss and perform experiment to show how sand can be obtained from water	RE star page 32 Sand Water Beaker RE star page 33 Filter paper Funnel	

		under the guidance of the teacher and then after learners presents their findings by writing on the chalk board. Each learner is tasked to write the notes in the exercise books using the relevant textbooks	Sand Water Beakers	
6.1	Explain how you can obtain salt from salty water by evaporation	- teacher briefly explains the term evaporation - Using teacher's explanation and the relevant textbooks Learners in groups discuss and perform experiment to show how salt can be obtained salty water under the guidance of the teacher and then after learners presents their findings by writing on the chalk board. - Each learner is tasked to write the notes in the exercise books using the relevant textbooks	RE star page 34-35 Heat source Salty water Evaporating dish Stirring rod Beakers	
6.2	know how to identify substances and their purity by using their melting and boiling point	- Defines the key terms purity, melting point and boiling point and demonstrates an experiment on the boiling point of water. - Learners in groups carry out experiments to determine melting point of naphthalene and boiling point of ethanol. - Learner writes a report on their findings.	Baroque page 25-26 Thermometer Stirrer Boiling tube Retort stand Heat source Triple stand Wire gauze Naphthalene Ethanol Water Safety tube Beaker Cork	
6.3	Ability to exhibit knowledge , skills and	Gives and guides learners on the activity of integration	RE; Baroque Pg 10,Star Pg 9	

Activity of integration	application of the key concepts on the experiment chemistry	Learner attempts the activity		
Theme: states and changes of states of matter. Competency: The learner uses knowledge of the arrangement and motion of particles to explain the properties of solids, liquids and gases.				
7.1 states and changes of states of Matter	- Define matter - State examples of matter - Identify the states of matter - Show that a stone has weight and occupies space	- Briefly explains the term matter using surrounding objects - Using teacher's explanation and the relevant textbooks Learners discuss in their groups to: - Define matter. - State examples of matter - Identify the states of matter - Show that a stone has weight and occupies space using the given material with the help of a teacher Learners are tasked to write their own notes in exercise books using the relevant textbooks	RE star page 39-40 Balloons filled with air Water Stone	
7.2 states and changes of states of Matter	Identifying states of matter using their properties.	- Explains and illustrates the arrangement of particles in matter using a chart. - Using teacher's explanation and the relevant textbooks Learners are tasked to draw diagram and explain the arrangement of particles of matter in their own words under the guidance of a teacher - Learners are tasked to write their own notes in exercise books using the relevant textbooks individually - Teacher explains and illustrates. - Using teacher's explanation and the relevant textbooks, learners in groups perform	RE star page 42 Chart showing arrangement of particles Textbooks Videos. RE star page 45	

	Demonstrate the space and movements of particles in matter	experiment and explains the space and movement of particle in matter under the guidance of a teacher Each learner is tasked to explain the movement of particles.	Marbles/small balls/ round seeds Flat tray Textbooks	
7.3	Demonstrates particles in gases using diffusion of perfumes vapour Demonstrates particles in liquids using potassium permanganate	- Teacher explains and illustrates. - Using teacher's explanation and the relevant textbooks, learners in groups perform experiment and explains the movement of particles in gases under the guidance of a teacher - Each learner is tasked to explain the movement of particles - Teacher explains and illustrates. - Using teacher's explanation and the relevant textbooks, learners in groups perform experiment and explains the movement of particles in liquids under the guidance of a teacher - Each learner is tasked to explain the movement of particles	RE star page 47 containers of perfume textbooks RE star page 48 Potassium permanganate Water Beakers Blue ink	
8.1	Explains the changes of state that occur when ice is heated	- Teacher briefly explains and illustrates. - Using teacher's explanation and the relevant textbooks, learners in groups perform experiment and explains the change of state that occurs when ice is heated under the guidance of a teacher - Each learner is tasked to write the notes in exercise book using the relevant textbooks	RE star page 49-52 Heat source Thermometer Beakers Ice cubes Stirring rod Stop watch or wall clock	

	Explains what happens to temperature of a solid as it is heated and cooled again	Using teacher's explanation and the relevant textbooks, learners in groups perform experiment and explains what happens to temperature of a solid as it is heated and cooled again and also Explains what happens to the temperature of a liquid as it cools under the guidance of a teacher Each learner is tasked to write the notes in exercise book using the relevant textbooks	RE star page 53 Beaker containing ice Source of heat Ice thermometer	
8.2	Explains what happens to temperature of a solid as it is heated and cooled again Explains what happens to the temperature of a liquid as it cools	- Teacher briefly explains and illustrates. - Using teacher's explanation and the relevant textbooks, learners in groups perform experiment and explains what happens to temperature of a solid as it is heated and cooled again and also Explains what happens to the temperature of a liquid as it cools under the guidance of a teacher - Each learner is tasked to write the notes in exercise book using the relevant textbooks	RE star page 53 Beaker containing ice Source of heat Ice thermometer	
8.3 Activity of integration	Ability to exhibit knowledge , skills and application of the key concepts on the states and changes of states of matter	- Gives and guides learners on the activity of integration - Learner attempts the activity	RE star page 54	
Theme: particle nature of matter Competency: the learners explore how materials are used and relates these uses to molecular structure				
9.1 Using materials	Know that materials used in everyday life can be classified in to natural and synthetic group and how this affects their use in everyday life.	- Teacher briefly explains on the materials used by human being and show some of the local materials used in everyday life. - Using teacher's explanation and the relevant textbooks Learners in groups	RE star page 57 Textbooks Some natural and artificial materials.	

		discuss to generate more materials and also the properties of materials given to them under the guidance of a teacher Learners are given a task to classify materials using their properties.		
9.2	Identify materials that conducts electricity and why materials conduct electricity - Comparing the tensile strength of different building materials - Find out what happens when different materials are hammered	- Teacher briefly explains why materials conduct electricity - Using teacher's explanation and the relevant textbooks Learners in groups discuss to Identify materials that conducts electricity under the guidance of a teacher and then after they present by writing on the chalk board - Learners write notes in their exercise books using the relevant textbooks	RE star page 59 Dry cell Electric wire and small bulb Piece of iron nails Piece of plastics Piece of wool Paper and glass Cotton cloth Rubber RE star Page 60 Fired bricks Unfired bricks Concrete blocks	

9.3	Explain the properties of materials in relation to their structures for example properties of glass that makes it suitable for different functions	- Teacher briefly explains the properties in relation to their structure. - Using teacher's explanation and the relevant textbooks Learners in groups discuss to Explain the properties of materials in relation to their structures under the guidance of a teacher and then after they present by writing on the chalk board - Learners write notes in their exercise books using the relevant textbooks	RE star page 64 Different glasses Textbooks	
10.1	- Explain how the structure of concrete influences the use of the concrete - Explain the effect of material ratio on strength of concrete	- Teacher briefly explains what we mean by a concrete - Using teacher's explanation and the relevant textbooks Learners in groups discuss to Explain how the structure of concrete influences the use of the concrete and the effect of material ratio on strength of concrete under the guidance of a teacher and then after they present by writing on the chalk board - Learners write notes in their exercise books using the relevant textbooks	RE star page 65 Magnifying glass/ hand lens 2 different pieces of concrete Concrete blocks Piece of culvert Concrete flower pot Concrete slab Paver pieces hammer	
10.2	- Define polymers - Classify materials as natural polymer or artificial polymers	- Teacher briefly explains substances which are made of long chain molecules - Using teacher's explanation and the relevant textbooks Learners in groups discuss to Define polymers and also Classify materials as natural polymer or artificial polymers under the guidance of a teacher and then after they present by writing on the chalk board	RE star page 67-68 Cotton wool Different plastics Water pipe	

[illegible]

	Identify natural and artificial fibres and state their uses	- Learners write notes in their exercise books using the relevant textbook - Teacher briefly explains the term fibre and shows fibres - Using teacher's explanation and the relevant textbooks learners in group discuss to identify natural and artificial fibres and state their uses under the guidance of teacher and then after they present by writing on the chalkboard - Learners write notes in their exercise books using the relevant textbook	RE star page 73-74 Textbooks Sisal Cotton Wool Silk Nylon	
11.2	Describe how they can dispose plastic materials at homes and also explain ways by which they can solve the environmental disposal problems of plastics	- Teacher briefly explains how plastics are disposed at school - Using teacher's explanation and the relevant textbooks learners in group discuss to describe how they can dispose plastic materials at homes and also explain ways by which they can solve the environmental disposal problems of plastics under the guidance of teacher and then after they present by writing on the chalkboard - Learners write notes in their exercise books using the relevant textbook	RE star page 76 textbooks	

11.3	Identify the materials That are: - Recycled - Non recycled - And state importance of recycling materials	- Teacher explains recycled and non-recycled materials - Using teacher's explanation and the relevant textbooks Learners in their group discuss and classify the materials that can be recycled and those that cannot be recycled under the guidance of teacher and then after they present by writing on the chalkboard - Learners are tasked to identify materials that can cause pollution to the environment.	RE star page 76-78 Textbooks Recycled and Non Recycled materials	
12.1 Activity of integration	Ability to exhibit knowledge , skills and application of the key concepts on using materials	- Gives and guides learners on the activity of integration - Learner attempts the activity	RE star page 81	

SUBJECT: Chemistry Class: S.1 Term: TWO

Teacher's Name:

Time allocation: 3 periods

When it comes to planning your lessons use the structure: starter, I do, we do, you do, plenary. The starter is a written task that reviews previous learning. Ensure your lessons provide regular and extended opportunities for independent practice.

YPR:

Y= yes, I taught the lesson

P= I partially taught it e.g. I didn't get through all the content.

R= I taught the lesson, but I think students would benefit from a review

Week Subtopic:	Learning outcomes	Methodology	Teaching materials	Y P R
Theme: Temporary and permanent changes to materials				
Competency: The learner recognises occurrence of temporary and permanent changes and their importance in everyday life				
1.1 Introduction	Learners should be able to ; - Define a change - Identify and explain the different changes in our daily life	- Teacher briefly explains using a chart with some example in our daily life - Using teacher's explanation and the relevant textbooks, learners in groups discuss to identify the different changes in our daily life under the guidance of a teacher. Learners present their findings by writing on chalk board. - Each learner write down the notes in their exercise books using the relevant textbooks.	- RE; Star Pg 85 - Chart with some examples showing changes	
1.2 Temporary and permanent changes	Learners should be able to ; show temporary (physical) changes	- Teacher briefly explains using a chart with some example in our daily life - Using teacher's explanation and the relevant textbooks, learners in groups perform an experiments to show temporary changes. Under the guidance of a teacher. learners present their finding by writing on the chalk board - Each learner write down the notes in their exercise books using the relevant textbooks	- RE; Star Pg 86 - Heat source - A piece of fat - Ice cold water - Ice bath - A hot saucer/ frying pan - Metallic cup	
1.3 Temporary and permanent changes	Learners should be able to ; show permanent (chemical) changes	- Teacher briefly explains using a chart with some example in our daily life - Using teacher's explanation and the relevant textbooks, learners in groups perform an experiments to show permanent changes. Under the guidance of a teacher. learners present their finding by writing on the chalk board - Each learner write down the notes in their exercise books using the relevant textbooks	- RE; Star Pg 87 - Source of heat - Wood splits - Piece of paper	

2.1 Temporary and permanent changes	Learners should be able to: Classify temporary and permanent changes using different materials	- The teacher demonstrates and explains by burning some substances in air i.e. melting candle wax, heating papers - using the teacher's demonstration and the relevant text book learners in groups discuss and perform experiment to classify temporary and permanent changes. Under the guidance of a teacher. learners present their finding by writing on the chalk board - Each learner write down the notes in their exercise books using the relevant textbooks.	- RE; Star Pg 88 - Textbooks - Heat source - Candle wax - Papers - Cloth - wood	
2.2 Temporary and permanent change	Learners should be able to: - Define A. temporary (physical) change B. Permanent (chemical) change - State examples of: A. Temporary B. Permanent	- The teacher demonstrates by burning some substances in air e.g. paper, cloth, wood - using the teacher's demonstration and the relevant text book learners in groups discuss and perform experiment to come up with the definitions of temporary and permanent change. They also identify examples of temporary and permanent changes. Under the guidance of a teacher. learners present their finding by writing on the chalk board - Learners are tasked to write their own definitions and examples in the exercise books using relevant text book	- RE; Star Pg 89 -90 - Textbooks - Paper - Cloth - Wood - Ice - Heat source	
2.3 Temporary and permanent changes	Identify and explain the examples of physical and chemical changes	- The teacher demonstrates by burning some substances in air e.g. paper, cloth, wood - Using teacher's explanation and the relevant textbooks Learners in groups perform experiments and discuss to identify and explain the examples of physical and chemical changes under the guidance of a teacher and learners	- RE; Star Pg89 - 90 - Textbooks - Paper - Cloth - Wood - Ice - Heat source	

		presents their finding to the class by writing on the chalk board. Learners are tasked to write the examples of physical and chemical change in their exercise books using the relevant text book.		
3.1 Temporary and permanent changes	Identify and explain the external conditions for any change to occur	- Teacher explains briefly by reviewing the previous lesson. - Using teacher's explanation and the relevant textbooks Learners in groups discuss to identify and explains the external conditions for any change to occur with the guidance of the teacher and learner present their findings. - Each learner write down the notes in their exercise books using the relevant textbooks	- RE; Star Pg 90 - Textbooks	
3.2 Temporary and permanent changes	To explain the differences between temporary and permanent change	- The teacher demonstrates by burning paper to ash and melting ice to water. - From the demonstration and the relevant text book, the learners discuss the differences between temporary and permanent changes. Under the guidance of a teacher. learners present their finding by writing on the chalk board - Learners are tasked to writer their own notes on the differences according to their own understanding using relevant text book	- RE star page 92 - Heat source - Match box - Paper - Ice - Candle wax	
Theme: Temporary and permanent changes of materials Competency: The learner recognises the characteristics of mixtures and compounds				
3.3 Pure and impure substances	- Define pure and impure substances - Classify pure and impure substances	- The teacher demonstrates the lesson by having pure water and impure water with sand.	- RE; Star Pg 97 - Pure water - Impure water - Beakers	

		- Learners use the demonstration to define pure and impure substances and also carry out an experiment to determine whether a substance is pure or not and classify them under the guidance of the teacher. Learners present their findings by writing on the chalk board - Learners are given a task to identify other pure and impure substances and write them in their exercise books.	- Heat source - Thermometer - Aluminium foil - Air - River water - Apple juice - Sugar solution - Milk - Cake and ice	
4.1 Pure and impure substances	Determine the boiling point of different substances	- The teacher briefly explains the boiling point substances for example the boiling point of ethanol is 78°C - Using the knowledge from the teacher and relevant textbooks, Learners in groups discuss to define a) boiling point b) melting point and perform experiment to determine the boiling point of water under the guidance of the teacher. Learners present their findings by writing on the chalk board - Learners write down the definition of boiling and melting point in their own words in their books using the relevant textbooks	- RE Prototype page 63 - Heat source - Beaker - Water - Ethanol - Thermometer	
4.2 Pure and impure substances	- Define freezing point - To compare the freezing point of water from different sources	- Teacher review and briefly explain about freezing. - Using teacher's explanation and the relevant textbooks, learners in groups discuss the definition of freezing and also compare the freezing point of water from different sources under the guidance of teacher. learners present their findings by writing on the chalk board - Learners write down the definition of freezing point and the comparison of the freezing point	- RE Star page 100 - Thermometer - Freezing mixture - Water bath - Retort stand 4 beakers - Water 3 boiling tubes	

		of water in different sources using the relevant text book		
4.3 elements	- Define element - State the examples of elements and their symbols.	- The teacher explains briefly about elements using the common materials made of elements like water which contains hydrogen and oxygen and salt which contains sodium and chlorine. - Using teacher's explanation and the relevant textbooks, Learners in groups brainstorm to define an element and also discuss to give examples of elements and their symbols under the guidance of teacher. learners present their findings by writing on the chalk board - Learners are tasked to write the definition of elements in their own words and state examples of elements using the relevant text book	- RE star page 101 - Water - Salt - Charts - Periodic table - Books	
5.1 elements	- Classify elements i.e. metals and non-metals - State the properties of metals and non-metals	- The teacher demonstrates by showing the class a metal i.e. iron and a non-metal i.e. sulphur. - Using teacher's explanation and the relevant textbooks, Learners in groups classify the elements as metals and non-metals and also state the properties of metals and non-metals under the guidance of teacher. learners present their findings by writing on the chalk board - Learners group metals and non-metals using the periodic table and write them in their exerciser books using the relevant textbooks.	- RE star page 103 - Periodic table - Charts with a lists of metals and non-metals - Books - Metal - Non-metals	

5.2 compounds	- Define a compound - State examples of compounds	- The teacher explains briefly about compounds using common substances e.g. water, salt and sugar - Using teacher's explanation and the relevant textbooks, Learners in groups discuss and come up with a definition of compounds and state examples of compounds metals under the guidance of teacher. learners present their findings by writing on the chalk board - After a discussion as a class, the learners should write their own definition of compounds and states examples in their exercise books using the relevant textbooks	- RE star page 103 - 104 - Water - Salt - Sugar - Periodic table - Charts with some examples of compounds	
5.3 compounds	To identify and name the elements in a compounds	- Teacher briefly explain how to identify elements from a compound using the chart - Using teacher's explanation and the relevant textbooks, Learners in groups to identify and name the elements in a compounds under the guidance of a teacher and then after they present their finding by writing on the chalk board. - Learners are given tasks to do in exercise books individually.	- RE star page 104 - Chart with some examples of compounds	
6.1 compounds	To classify the substances as elements and compounds	- Teacher reviews and briefly explains using substances in our daily life - Using teacher's explanation and the relevant textbooks, Learners in groups to classify the substances as elements and compounds under the guidance of teacher and they present their findings by writing on chalk board - Learners are given tasks to do in exercise books individually.	- RE star page 104 - Chart with locally available substances - Sugar - Common salt - Gold jewellery - Aluminium foil - Glass - Iron rod - Water	

			charcoal	
6.2 compounds	To explain how a compound is formed for example making a compound from iron and sulphur	- Teacher briefly demonstrates for example burning magnesium in air to form magnesium oxide - Using teacher's demonstration and the relevant textbooks, learners in groups discuss the making of a compound from iron and sulphur under the guidance of a teacher and after they present their finding by writing on chalk board - Learners are given tasks to do in exercise books individually.	- RE star page 105 - Iron filings - Heat source - Powdered sulphur - Test tubes	
6.3 compounds	- Explain the effect of heating a mixture and iron and sulphur - Identify the product formed - Write the word equation for reaction	- The teacher explains and demonstrates by heating a mixture of iron and sulphur - From the teacher's demonstration and using relevant textbooks, learners discuss and in groups to be able to: - Note their observations of the reaction - Name the product - Write the word equation for the reaction observed under the guidance of a teacher and after they present their finding by writing on chalk board - After discussion as a class where groups share their ideas, learners are instructed to write their own notes in their exercise books using the relevant textbooks.	- RE star page 105 - Heat source - Test tube - Sulphur - Iron - Textbooks	
7.1 Mixtures,	- Define a mixture - Give examples if mixtures	- The teacher explains and demonstrates by creating a mixture of sand and rice. - Using teacher's explanation and the relevant textbooks learners in groups brainstorm and	- RE star page 106 - Rice - Sand - Water	

		define the term “mixture” and also discuss to think about some examples of mixtures under the guidance of a teacher and after learners present their finding on the chalk board. Learners are tasked to then write their own definition and examples of mixtures in their exercise books.	- Charts with many substances - Beakers - Books	
7.2 Mixtures,	Explain the types of mixtures and their examples	- The teacher demonstrates by having two mixtures i.e. homogenous mixture and heterogeneous mixture. - From the teacher’s demonstration and using relevant textbooks, Learners in their respective discuss and explain the different types of mixtures and suggest examples of each type Under the guidance of a teacher and learners present their findings by writing on the chalk board - After discussion as a class where groups share their ideas, learners are instructed to write their own notes in their exercise books using the relevant textbooks.	- RE star page 109 - Oil - Water - Ethanol - Separating funnel - Glass beakers - Boiling tubes	
7.3 Mixtures,	To state the differences between mixtures and compounds	- Teacher explains and demonstrates by separating the following; (a) Sulphur and iron (b) Iron (ii) sulphide One of them will not be separated indicating that one of the substance is a mixture and another is a compound - Using teachers explanation and the relevant textbooks, learners in groups discuss to come up with the differences between mixtures and compounds under the guidance of the teacher and after they present their findings by writing on the chalk board	- RE star page 111 - Iron (ii) sulphide - Sulphur and iron - Magnet - Textbooks	

		Learners write notes in their exercise books using the relevant textbooks		
8.1 Mixtures,	- Identify a method of separating mixtures - Understand the principle of separation.	- The teacher demonstrates and explains on method of separating mixtures. - Using teacher's explanation and the relevant textbooks Learners discuss in their groups the method of separating mixtures and state the principle of separation under the guidance of a teacher and after learners present their finding on the chalk board. - Learners make notes in their exercise books on the principle of separation and the method using relevant textbooks	- RE Prototype page 71 - Beaker - Funnel - Filter paper - Books - Charts	
8.2 Mixtures,	- Define decanting - Identify examples of mixtures that can be separated by decanting - Explain how decanting is used in the separation of mixtures	- The teacher briefly explains the method of decanting. - Using teacher's explanation and the relevant textbooks Learners in their groups discuss and then perform the decanting experiment. They should then discuss and agree: - Definition of decanting - Examples of mixtures that can be separated by decanting - An explanation on how decanting is used in the separation of mixtures under the guidance of a teacher and after learners present their finding on the chalk board. - Each group shares their ideas and then learners are instructed to write their own notes in their exercise books on the above using relevant textbooks	- Prototype page 71 - Sand - Water - Empty beakers	

8.3 Mixtures,	- Define filtration - Identify examples of mixtures that can be separated by filtration - Explain how filtration is used in the separation of mixtures	- The teacher briefly explains the method of filtration. - Using teacher's explanation and the relevant textbooks In groups, learners perform an filtration experiment to be able to discuss and agree: - A definition of filtration - Examples of mixtures that can be separated by filtration - An explanation on how filtration is used in the separation of mixtures. Under the guidance of a teacher and after learners presents their finding on the chalk board. - Learners are instructed to write their own notes in their exercise books using relevant textbooks	- RE Prototype page 71 - Water - Sand - Funnel - Filter paper - Empty beakers	
9.1 Mixtures,	- Define distillation - Identify examples of mixtures that can be separated by distillation - Explain how distillation is used in the separation of mixture	- The teacher briefly explains the method. - Using teacher's explanation and the relevant textbooks , Learners in their groups discuss and perform an experiment to come up with the: - Definition of distillation - Examples of mixtures that can be separated by distillation - Explanation on how distillation is used in the separation of mixtures. Under the guidance of a teacher and after learners presents their finding on the chalk board. - Learners are tasked to write their own notes in their exercise books on the above using relevant textbooks	- RE Prototype page 71 - Heat source - Distillation flask - Dirty water - Empty conical flasks - Beakers - Thermometer	
9.2 Mixtures,	- Define fractional distillation. - Identify examples of mixtures that can be	- The teacher briefly explains the methods by showing the learners liquids with different boiling points. - Using teacher's explanation and the relevant textbooks In groups, learners discuss to come up	- RE star page 113 - Water - Ethanol - Thermometer - Heat source	

	separated by fractional distillation. Explain how fractional distillation is used in the separation of mixture.	with a definition of fractional distillation and also identify examples of substances that can be separated by distillation under the guidance of a teacher and after learners present their finding on the chalk board. Learners then write definitions in their exercise books in their own words using relevant textbooks	- Empty conical flasks - Beakers	
9.3 Mixtures,	- Define evaporation - Identify examples of mixtures that can be separated by evaporation - Explain how evaporation is used in the separation of mixtures	- The teacher briefly explains the methods. - Using teacher's explanation and the relevant textbooks , Learners in groups discuss and perform an experiment to be able to: a) Definition of evaporation b) Examples of mixtures that can be separated by evaporation - Explanation on how evaporation is used in the separation of mixtures. Under the guidance of a teacher and after learners present their finding on the chalk board. - Learners are tasked to write notes on the above according to their own understanding using relevant textbooks	- RE Prototype page 71 - Water - Salt - Heat source - Evaporation dish - Empty beakers	
10.1 Mixtures,	- Define chromatography - Identify examples of mixtures that can be separated by chromatography - Explain how chromatography is used in the separation of mixtures	- The teacher briefly explains the method. - Using teacher's explanation and the relevant textbooks , Learners in groups discuss and perform an experiment to come up with: a) A definition of chromatography b) Examples of mixtures that can be separated by chromatography c) Explanation of how chromatography is used in the separation of mixtures	- RE star page 108 - Filter paper - Solvent e.g. water - Beakers - Ink	

10.2 Mixtures,	• Define sublimation • Identify examples of mixtures that can be separated by sublimation • Explain how sublimation is used in the separation of mixtures	• The teacher briefly explains the methods. • Using teacher's explanation and the relevant textbooks Learners in groups discuss and perform an experiment to come up with: a) Definition of sublimation b) Examples of substances that can be separated by sublimation • Explanation on how sublimation is used in the separation of mixtures. Under the guidance of a teacher and after learners present their finding on the chalk board. • Learners are instructed to write notes on their own in their exercise books using relevant textbooks	• RE star page 116 • Iodine • Ammonium chloride • Iron (iii) chloride • Aluminium chloride • Sodium chloride • Heating source • Flask	
10.3 Mixtures,	• Define fractional crystallisation • Identify examples of mixtures that can be separated by fractional crystallisation.	• The teacher briefly explains the method. • Using teacher's explanation and the relevant textbooks Learners in groups discuss to be able to: a) Define fractional crystallisation • Give examples of substances that can be separated by fractional crystallisation. Under the guidance of a teacher and after learners present their finding on the chalk board. • Learners are tasked to write notes on their own in their exercise books using relevant textbooks	• RE Prototype page 71 • Mixture of iron and carbon • Mixture of iron and sulphur • Magnet • Paper	

11.1 Mixtures,	- Identify examples of mixtures that can be separated by a magnet - Explain how a magnet is used in the separation of mixtures	- The teacher briefly explains the method. - Using teacher's explanation and the relevant textbooks Learners in groups discuss to identify examples of mixtures that can be separated by magnets and give an explanation for the separation of mixtures using magnets. Under the guidance of a teacher and after learners present their finding on the chalk board. - Learners are tasked to write notes in their exercise books using relevant textbooks	- RE star page 117 - Mixture of Iron and carbon - Mixture of iron and sulphur - Magnet - Paper	
11.2 Mixtures,	- Identify examples of mixtures that can be separated by a separating funnel - Explain how the mixtures can be separated using the separating funnel	- Teacher briefly explains the method - Using teacher's explanation and the relevant textbooks Learners discuss to identify examples of mixtures that can be separated by a separating funnel and also give an explanation for the separation of mixtures using separating funnels. Under the guidance of a teacher and after learners present their finding on the chalk board. - Learners are tasked to write their own notes in their exercise books using relevant textbooks	- RE star page 115 - Oil - Water - Separating funnel - Empty beaker	
11.3 Mixtures,	- Define a) miscible liquids b) immiscible liquids - Identify examples of miscible and immiscible liquids	- The teacher briefly explains liquids - Using teacher's explanation and the relevant textbooks Learners in their groups discuss to be able to: a) Define 1) miscible liquids 2) immiscible liquids - Identify examples of miscible and immiscible liquids. Under the guidance of a teacher and after learners present their finding on the chalk board. - Learners are tasked to write their own notes in their exercise books using relevant textbooks	- RE star page 115 - Mixture of oil and water - Par affine and water - Ethanol and water - Spirit and water	

12 Activity of integration	Ability to exhibit knowledge , skills and application of the key concepts on the mixtures elements and compounds	Gives and guides learners on the activity of integration Learner attempts the activity	RE star page 118	
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SUBJECT: Chemistry Class: S.1

Term: THREE

Teacher's Name:

Time allocation: 3 periods

When it comes to planning your lessons use the structure: starter, I do, we do, you do, plenary. The starter is a written task that reviews previous learning. Ensure your lessons provide regular and extended opportunities for independent practice.

YPR:

Y= yes, I taught the lesson

P= I partially taught it e.g. I didn't get through all the content.

R= I taught the lesson, but I think students would benefit from a review

Week Subtopic:	Learning outcomes	Methodology	Teaching materials	Y P R
Theme: air and atmosphere				
Competency: The learners appreciate that air is a mixture of gases in which oxygen is the active constituent				
1.1 Air	- Define the term 'air' - State the composition of air - Describe an experiment to show that part of air is used	- The teacher introduces the concept by explaining the term atmosphere and stating the components of atmosphere. - Using teacher's explanation and the relevant textbooks Learners in their groups brainstorm the definition, state the composition of air and	- Baroque page 83-84 - Star page 122-123 - Textbooks - Trough - Water - Match box	

	up when the candle burns	describe an experiment to show that part of air is used up when the candle burns under the directives of a teacher, and after learners present their finding on the chalk board Learners are tasked to write their own notes in their exercise books using relevant textbooks	- Gas jars with a cork - candle	
1.2 Air	Describe an experiment to determine the percentage of oxygen in air using heated copper wire	- Teacher explains and displays the charts with a diagram for the experiment. - Using teacher's explanation and the relevant textbooks Learners in their groups describe how to determine the percentage of air under the guidance of the teacher and there after they present their findings by writing on the chalk board - Learners are tasked to write their own notes in their exercise books using relevant textbooks	- RE star page 124-125 - Charts with a diagram fir the experiment - Textbooks	
1.3	- State the method for separating components of air - Describe how air is separated by fractional distillation	- Explains and displays the chart with drawn diagram of fractionating column. - Using teacher's explanation and the relevant textbooks Learners in their groups state the methods of separating air and describe how air is separated by fraction distillation under the guidance of the teacher and there after they present their findings by writing on the chalk board - Learners write down the steps for separating air by fractional distillation.	- RE star page 127-128 - Chart with drawn up diagram. - Textbooks	
2.1	- To state the uses of components of air - State how the components of air are a) increased b) reduced in the atmosphere	- Narrates a story about a person who fainted when he was running because of lack of oxygen. - Using teacher's explanation and the relevant textbooks Learners in groups to state the uses of components of air and also state how the components can be increased or reduced from the atmosphere under the guidance of the	- RE star page 129 - Textbooks	

		teacher and there after they present their findings by writing on the chalk board Learners write their own notes in their books using the relevant textbooks		
2.2 Pollution	- Define the term air pollution - State the sources of pollutants of air and explain the effects and measures to be taken to prevent air pollution	- Teacher narrates a story when he/she went in a school kitchen full of smoke and explained how it affected him /her. using teacher's explanation and the relevant textbooks Learners in groups and generate the definition of air pollution, State the sources of pollutants of air and explain the effects and measures to be taken to prevent air pollution under the guidance of the teacher and there after they present their findings by writing on the chalk board - Learners write their own notes in their books using the relevant textbooks	- RE star page 129-131 - Textbooks - Videos showing pollution of air	
2.3 Combustion	- Define the term combustion - Identify the examples of daily life combustion and active component in combustion and identify the products of combustion	- Teacher explains and demonstrates by burning a piece of paper in air. - Using teacher's explanation and the relevant textbooks Learners in groups generate the definition combustion, Identify the examples of daily life combustion and active component in combustion and identify the products of combustion under the guidance of the teacher and there after they present their findings by writing on the chalk board. - Learners write their own notes in their books using the relevant textbooks	- RE star page 132-134 - Two candles - Match box - A glass or metal cover and two beakers - Plastic water bottle - Lime water - Deflagrating spoon or stiff wire	

3.1 Combustion	- Define the term an oxide - Explain what happens to elements when they are burnt in air and write down the equation for the reaction	- Teacher explains and demonstrates how magnesium burns in air - Using teacher's explanation and the relevant textbooks Learners in groups discuss and perform experiments to generate the definition of an oxide, explain what happens to elements when they are burnt in air and write down the equation for the reaction under the guidance of the teacher and there after they present their findings by writing on the chalk board. - Learners write their own notes in their books using the relevant textbooks	- RE star page 134-136 - Textbooks - Heat source - Magnesium ribbon - Carbon - Phosphorus - sodium	
3.2 oxides	classify an oxide as acidic, basic, amphoteric mixed and neutral	- Teacher explains and demonstrates by dissolving some of the soluble oxides in water and then tests using litmus papers. - Using teacher's explanation and the relevant textbooks Learners in groups discuss and perform experiments to classify oxide as acidic, basic, amphoteric mixed and neutral under the guidance of the teacher and there after they present their findings by writing on the chalk board. - Learners write their own notes in their books using the relevant textbooks	- RE star page 135 - Magnesium ribbon - Carbon - Phosphorus - sodium	
3.3 rusting	- Define the term - Rust - Rusting - State the chemical name and formula of rust - Identify the conditions necessary for rusting	- Carries an iron tool which was left in the air for 3 days and explains what can be observed (rusting). - Using teacher's explanation and the relevant textbooks Learners in groups discuss to generate the definition of rust, rusting and state the chemical name and formula of rust neutral under the guidance of the teacher and there after they present their findings by writing on the chalk board.	Panga or tool coated with rust	

		Learners are tasked to write down the conditions of rusting.		
4.1 rusting	Describe an experiment to investigate how oxygen and water are involved in rusting.	- Explains the already prepared experiment - Using teacher's explanation and the relevant textbooks Learners in groups discuss to identify which conditions were missing in the experiment. - Learners write down the steps involved in the preparation on their own using the relevant textbooks	- RE star page 140 - Open container containing rusted iron nails with cold water - Closed container with un-rusted iron nails and boiled water	
4.2 rusting	- To state the ways of reducing iron rusting - Give the disadvantages of rusting	- Illustrates how oil can limit oxygen to react with iron nails in the presence of water. - Using teacher's explanation and the relevant textbooks Learners in groups discuss to state ways of preventing iron from rusting how rusting and the disadvantages of rusting under the guidance of the teacher and there after they present their findings by writing on the chalk board. - Learners write their own notes in their exercise books.	- RE star page 141 - Textbooks	
4.3 Activity of integration	Ability to exhibit knowledge , skills and application of the key concepts on air	- Gives and guides learners on the activity of integration - Learner attempts the activity	RE star page 142	
Theme: Air and environment				
Competency: The learner investigates evaporation and condensation as a natural process essential for replenishing supplies of fresh water in the cycle				
5.1 Water	- Name sources of water - State the impurities found in water	- Teacher takes learners to some sources of water near the school <i>Note should be done before the lesson</i> - Using the knowledge from the tour and the relevant text book Learners in their groups	- RE star page 147-148 - Text books - Sources of water - Learners books	

		discuss to identify different sources of natural water. Learners then conduct an experiment to identify impurities in the water under the guidance of the teacher and after learners present their findings Learners write their own notes and observations after a class discussion using the relevant textbooks		
5.2 Water	Explain the major uses of water in everyday life	- Teacher explains some uses of water at homes - Using teacher's explanation and the relevant textbooks Learners in groups discuss the major uses of water under the guidance of the teacher and after learners present their findings - Learners write their own notes in their exercise books using the relevant textbooks	- RE star page 150 - Textbooks	
5.3 Water	Draw and explain the water cycle	- Teacher briefly explain how water is formed - Using teacher's explanation and the relevant textbooks Learners in groups discuss to draw and explain the water cycle under the guidance of the teacher and there after learners presents their findings by writing on the chalk board - Learners write their own notes in their exercise books using the relevant textbooks	- RE star page 150-151 - Textbooks - Chart show water cycle	
6.1 Water	Explain the physical properties of water	- Teacher briefly explains what is meant by physical properties - Using teacher's explanation and the relevant textbooks Learners in groups discuss to explain the physical properties of water under the guidance of a teacher and there after they presents their findings by writing on the chalk board - Learners write their own notes in their exercise books using the relevant textbooks.	- RE star page 152-154 - Textbooks	

6.2 Water	Explains the chemical properties of water	- Teacher explains the effect of water on different substances - Using teacher's explanation and the relevant textbooks Learners in groups discuss to explain the chemical properties of water ie the reaction of water with sodium and potassium under the guidance of a teacher and there after they presents their findings by writing on the chalk board - Learners write their own notes in their exercise books using the relevant textbooks.	- RE star page 154 - Baroque page 98 - Text book - Water - Potassium and sodium - beakers	
6.3 Water	explain the chemical tests of water	- Teacher narrates a story when he sent a boy to bring some water from the lab. As he did not test instead he brought acid. - Using teacher's explanation and the relevant textbooks Learners in groups discuss to explain the chemical tests for water under the guidance of a teacher and there after they presents their findings by writing on the chalk board - Learners write their own notes in their exercise books using the relevant textbooks	- RE star page 156 - Water anhydrous copper(ii) sulphate - Blue cobalt (ii) chloride paper	
7.1 Water purification	- Define water purification - Identify the methods used in water purification	- Recaps on the characteristics of pure substances for learners to be able to define water purification. - Using the knowledge from the teacher, Learners discuss in their groups to generate the definition and methods used to purify water under the guidance of the teacher - Learners are given a task to present in their findings and write their own notes using the relevant textbooks	- RE star page 157 - Text book - Chlorine tablets - Water guard	

7.2 Water pollution	- To define water pollution - State the pollutants in water - Describe the ways of reducing water pollution.	- Teacher talks about those who reside in places without toilets and how they end up polluting water with their waste. - Using the knowledge from the teacher and the relevant textbooks, Learners discuss in their groups to generate the definition of pollution, identify the pollutants in water and also describe the ways of reducing water pollution under the guidance of the teacher and after they present their findings by writing on the chalk board - Learners are asked to write down their notes about water pollution and how it is avoided using the relevant textbooks.	- RE star page 158 - Textbooks - Learners books	
7.3 Water pollution and sewage	- Define the term sewage - Describe the steps taking in sewage treatment	- Teacher narrates a story when he went around lake shores and found people disposing rubbish faces in the lake and yet it is the water they normally use at home. - Using the knowledge from the teacher, Learners discuss in their groups to generate the definition of sewage, and describe the steps taking in sewage treatment under the guidance of the teacher and after they present their findings by writing on the chalk board - Learners write their own notes in their exercise books using the relevant textbooks	- RE star page 158-159 - Textbooks - Learners books	
8.1 Activity of integration	Ability to exhibit knowledge , skills and application of the key concepts water	- Gives and guides learners on the activity of integration - Learner attempts the activity	RE star page 160	
Theme: rocks and minerals Competency: The learners appreciate that rocks are composed of different minerals and these determine the properties of the rocks				

8.2 Rocks	- Define the term rocks - Identify the physical properties of rocks - Identify characteristic differences between rocks	- Teacher takes learners to tour in mountainous area to see different rocks - Using the knowledge from the tour and the relevant textbooks, learners in groups brainstorm to define a rock, identify the physical properties of rocks and also identify the characteristic differences between rocks under the guidance of the teacher and after they present their findings by writing on the chalk board - Learners write their own notes in their exercise books using the relevant textbooks	- RE star page 163-166 - Textbooks	
8.3 rocks	Explain the different types of rocks, how they form and how they change	- Teacher explains by showing learners different rocks - Using the knowledge from the teacher and the relevant textbooks, Learners discuss to explain the different types of rocks, how they form and how they change under the guidance of the teacher and after they present their findings by writing on the chalk board - Learners write their own notes in their exercise books using the relevant textbooks	- RE star page 166 - Textbooks	
9.1 minerals	- Define the term mineral - Identify examples of minerals - State the characteristics of minerals	- Teacher explains by showing some examples of mineral - Using teacher's explanation and the relevant textbooks, Learners look at pictures in their books, and discuss to define 'minerals' and identify examples of minerals under the guidance of the teacher and after they present their findings by writing on the chalk board - Learners write their own notes in their exercise books using the relevant textbooks	- RE star page 167-170 - Textbooks - Learner books - Pictures of minerals	

9.2 minerals	• Use the characteristics of minerals to identify the uses of the mineral • Identify the minerals which are: a) Hard b) soft	• Gives a brief explanation about hard and soft minerals. • Using teacher's explanation and the relevant textbooks, Learners discuss in groups to generate the uses of minerals and identify the mineral which are hard and soft under the guidance of the teacher and after they present their findings by writing on the chalk board • Learners write their own notes. Using the relevant textbooks	• RE star page 169 • Textbooks • Learners books	
9.3 weathering	• Explain what is meant by physical weathering • Explain how rocks are broken down by freezing and thawing • Demonstrate physical weathering by the process of abrasion.	• Teacher explain by showing learners the rock which was broken down by physical weathering • Using teacher's explanation and the relevant textbooks, Learners discuss in groups to explain what is meant by physical weathering, explain how rocks are broken down by freezing and thawing and demonstrate physical weathering by the process of abrasion under the guidance of the teacher and after they present their findings by writing on the chalk board • Learners write their own notes. Using the relevant textbooks	• RE star page 173 • Textbooks	
10.1 weathering	• Explain how rocks are broken down by; (a) plants (b) animals	• Teacher explain by showing learners the rock which was broken down by man • Using teacher's explanation and the relevant textbooks, Learners discuss in groups to explain how rocks are broken down by plants and animals under the guidance of the teacher and after they present their findings by writing on the chalk board • Learners write their own notes. Using the relevant textbooks	• RE star page 174 • Textbooks • stones	

10.2 Weathering	- define chemical weathering - explain the effects of chemical weathering on rocks	- Teacher explain by showing learners the rock which was broken down by chemical weathering - Using teacher's explanation and the relevant textbooks, Learners discuss in groups to define chemical weathering and explain the effects of chemical weathering on rocks under the guidance of the teacher and after they present their findings by writing on the chalk board - Learners write their own notes. Using the relevant textbooks	- RE star page 174-175 - Textbooks - stones	
10.3 weathering	- demonstrates and explains the weathering by; (a) water (b) acidic water	- Teacher explain by showing learners the rock which was broken down by acidic water - Using teacher's explanation and the relevant textbooks, Learners discuss in groups to explain how rocks are broken down by water and acidic water under the guidance of the teacher and after they present their findings by writing on the chalk board - Learners write their own notes. Using the relevant textbooks	- RE star page 176-178 - Textbooks - Broken rocks	
11.1 Soil formation	- Name the components of soil - Identify the layers of the soil - Make the layer of soil	- Teacher explain by showing learners the soil components - Using teacher's explanation and the relevant textbooks, Learners discuss in groups to Name the components of soil, identify the layers of the soil And make the layer of soil under the guidance of the teacher and after they present their findings by writing on the chalk board - Learners write their own notes. Using the relevant textbooks	- RE star page 179-181 - Textbooks - Soil sample	

11.2 Activity of integration	Ability to exhibit knowledge , skills and application of the key concepts on mineral and rocks	- Gives and guides learners on the activity of integration - Learner attempts the activity	RE star page 183	
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Note 1. Each is a single lesson.

2. Term one schemed for 14 weeks following directives from the ministry of education